

IIITH PGEEE 2021

1. No of ones in binary representation of $(3 \times 2048 + 15 \times 512 + 5 \times 16 + 3)$.
2. Charact. Equation of SR notch using NOR gates.
3. One question of Calendar.
4. What Bakery Algorithm improves: (a). throughput (b). latency (c). both (d). None
5. "COMPUTER" total arrangements possible with first and last character consonants only.
6. Implementation of queue using linked list and array. (a) $O(1)$ to dequeue or enqueue using LL. (b) $O(1)$ to dequeue or enqueue using Array. (c). Both a and b (d) None.
7. One question on insertion of elements in Empty AVL tree.
8. One question on $h(k) = k \bmod 7$ (using Quadratic Probing) with values given. Guess the array after element insertion.
9. X% error in measuring side of a cube. What would be the % error in measuring surface area of cube.
10. 40ft tall tower. If a monkey climbs 5ft and slides 3ft. How much steps to reach at top of building.
11. One question from Number series.

12. (Only C language) Two coding questions with 8 test cases. 3 test cases were public, rest were private. (a). Print absolute mod of numbers. (b) Find total number of primes between 1 to ($\leq$) n.
13. POS equation was given and SOP was asked. (DL)
14. One Question from MUX(2 x 1), asked to implement A+B.
15. One question of LIKE operator SQL.
16. Few Questions from Spatial Aptitude.
17. Find the sorting algorithm.
18. Cloth: Fabric :: sound then (a)music (b)pitch (c)decibel (d)vibrations
19. 5 basic comprehensions were given. Each Comprehensions consists of 3-5 questions.
20. MSQ: Possible subnet mask of class A IPV4. (a)255.0.0.0 (b)255.255.0.0 (c)255.255.255.0 (d)255.255.255.255
21. MSQ: BST deletion of node with two children but distinct elements in BST (a). replace deleted node with minimum elem. of left subtree. (b) replace deleted node with minimum elem. of right subtree. (c) replace deleted node with max elem. of left subtree. (d) replace deleted node with max elem. of right subtree.
22. Minimum nor gates to implement $AB+C$.

- 23. A line segment AB is given of length $2a$ and point is marked on this line say "P". Find the probability such that the area of rectangle formed from AP and PB is $(a^2)/2$.
- 24. One question from Direct Mapping.
- 25. One question from Set Associative Mapping.
- 26. Definition of Optimal Page replacement Algo.
- 27. One from Logical data Independence.
- 28. ...

50 questions in Aptitude section and 42 questions in Technical Section.